Dominant Strategies and Nash Equilibrium

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Iterated Elimination

Today we'll look at games that cannot be solved with dominance alone.

We'll be using two tools in particular:

- **iterated elimination** and
- **Nash equilibrium**.

Associated Reading

Game Theory, Sections 2.5 and 2.6

Consider this game:

	L	C	R
T	(3, 1)	(0, 2)	(1, 3)
M	(1, 3)	(1, 1)	(2, 0)
B	(0, 2)	(2, 0)	(3, 1)

No row or column is strictly dominated by another single row or column.

But we can still make progress by thinking carefully about what rational players will rule out.

The procedure:

1. Remove any strictly dominated strategy for any player.
2. In the smaller game that remains, look for newly dominated strategies.
3. Repeat until no more eliminations are possible.

This is justified by **common knowledge of rationality**: if everyone is rational and knows everyone is rational, no one will play a dominated strategy, so we can treat the game as if those strategies don't exist.

	L	M	R
T	(4, 3)	(2, 1)	(3, 0)
B	(2, 1)	(1, 2)	(1, 3)

Step 1: Check Player 2's columns. Is R dominated?

- Under T: M gives 1, R gives 0. So M is better than R when Row plays T.
- Under B: M gives 2, R gives 3. So R is better than M when Row plays B.

R is not dominated. Let's check whether M or L dominate each other... actually, check if R is dominated by L:

- Under T: L gives 3, R gives 0 → L better.
- Under B: L gives 1, R gives 3 → R better.

No strict dominance for Player 2 here. Let's check Player 1.

	L	M	R
T	(4, 3)	(2, 1)	(3, 0)
B	(2, 1)	(1, 2)	(1, 3)

Check Player 1: Does T dominate B?

- Against L: T gives 4, B gives 2 → T better ?
- Against M: T gives 2, B gives 1 → T better ?
- Against R: T gives 3, B gives 1 → T better ?

T strictly dominates B. Eliminate B.

After Eliminating B

The game shrinks to one row:

	L	M	R
T	(4, 3)	(2, 1)	(3, 0)

Now Player 2 chooses between L (payoff 3), M (payoff 1), R (payoff 0).

L strictly dominates M and R. Player 2 chooses **L**.

Solution by iterated elimination: (T, L) with payoffs **(4, 3)**.

The Limits of Iterated Elimination

Iterated elimination works neatly when it resolves the game completely.

Often it doesn't. Many games survive iterated elimination with multiple strategy profiles remaining.

We need a solution concept that applies to any game.

Nash Equilibrium

A strategy profile is a **Nash equilibrium** if no player can improve their payoff by *unilaterally* changing their strategy (given that all other players keep theirs fixed).

Definition: A strategy profile $(s_1, s_2, \dots, s_n)$ is a Nash equilibrium if for every player i and every alternative strategy s_i':

$$\pi_i(s_i, s_{-i}) \geq \pi_i(s_i', s_{-i})$$

Equivalently: each player is playing a **best response** to what the others are doing.

	Ed: Normal	Ed: Extra
Doug: Normal	(2, 2)	(0, 3)
Doug: Extra	(3, 0)	(1, 1)

Is (Normal, Normal) a Nash equilibrium?

No, since Doug can improve from 2 to 3 by switching to Extra (given Ed plays Normal).

Is (Extra, Extra) a Nash equilibrium?

Yes, since given Ed plays Extra, Doug gets 1 from Extra vs 0 from Normal. No improvement. Same for Ed.

Nash Equilibrium in the Coordination Game

	Drive Left	Drive Right
Drive Left	(2, 2)	(0, 0)
Drive Right	(0, 0)	(2, 2)

(Left, Left): given P2 plays Left, P1 gets 2 from Left vs 0 from Right. No improvement.

(Right, Right): given P2 plays Right, P1 gets 2 from Right vs 0 from Left. No improvement.

Two Nash equilibria. Each is stable; the problem is coordination.

Nash Equilibrium in Battle of the Sexes

	Opera	Football
Opera	(2, 1)	(0, 0)
Football	(0, 0)	(1, 2)

(Opera, Opera): P1 gets 2; switch to Football gives 0. P2 gets 1; switch to Football gives 0. ?

(Football, Football): P1 gets 1; switch to Opera gives 0. P2 gets 2; switch to Opera gives 0. ?

Again two equilibria. This time each favors a different player.

Two hunters can cooperate to hunt a stag (high reward) or hunt a rabbit individually (safe reward).

	Hunt Stag	Hunt Rabbit
Hunt Stag	(4, 4)	(0, 3)
Hunt Rabbit	(3, 0)	(3, 3)

Is (Stag, Stag) a Nash equilibrium?

The Stag Hunt

Two hunters can cooperate to hunt a stag (high reward) or hunt a rabbit individually (safe reward).

	Hunt Stag	Hunt Rabbit
Hunt Stag	(4, 4)	(0, 3)
Hunt Rabbit	(3, 0)	(3, 3)

Given P2 hunts Stag, P1 gets 4 from Stag vs 3 from Rabbit.

Is (Rabbit, Rabbit) a Nash equilibrium?

Given P2 hunts Rabbit, P1 gets 3 from Rabbit vs 0 from Stag.

Stag Hunt: Two Equilibria, Different Characters

	Hunt Stag	Hunt Rabbit
Hunt Stag	(4, 4)	(0, 3)
Hunt Rabbit	(3, 0)	(3, 3)

(Stag, Stag) is **Pareto superior**. Both do better than in (Rabbit, Rabbit).

But (Stag, Stag) is **riskier**: if your partner defects to Rabbit, you get 0.

(Rabbit, Rabbit) is **safer**: you get 3 no matter what the other player does.

Prisoner's Dilemma vs. Stag Hunt

Both games have a Pareto-superior cooperative outcome that's not always achieved.

But they're structurally different:

- In the **PD**, defection is a *dominant* strategy. Rational even if you know the other will cooperate.
- In the **Stag Hunt**, cooperation is rational *if you trust* the other player will cooperate. But unilaterally it's risky.
- In PD the problem is incentives. In Stag Hunt the problem is **trust and coordination**.

Real Life

What are some cases in your life that feel like PDs?

What are some cases that feel like Stag Hunts?

Nash equilibrium identifies **stable** outcomes: situations where no one has reason to deviate.

It does **not** tell us:

- Which equilibrium will be reached when there are multiple.
- That equilibria are *good*. The PD equilibrium is Pareto inferior.
- How players decide what to do when they haven't communicated.

Nash equilibrium is a *necessary condition* for a stable rational outcome, not a complete theory of rational play.

Three Kinds of Game Theory

Bonanno's book focuses on **rational choice game theory**: assume common knowledge of rationality, deduce what rational players will do.

But game theory is also done in two other ways.

Evolutionary game theory: strategies are hard-wired, and populations of strategies grow proportionally to their success. No assumption of rationality. Selection does the work.

Experimental game theory: put real people in labs and see what they actually do. Results often diverge from rational-choice predictions.

Why the Distinctions Matter

In the PD, rational-choice theory predicts (Defect, Defect).

Experimental evidence: people frequently cooperate, especially when they interact repeatedly. The Axelrod tournaments showed that “tit for tat” strategies do very well.

Evolutionary simulations: populations can evolve toward cooperation even without any rationality assumption.

All three approaches study the same games but ask different questions and reach different conclusions.

Worked Examples

A seafront promenade has seven numbered stalls (1 through 7, left to right). Two ice cream vendors choose a stall simultaneously. Beachgoers buy from the nearer vendor; ties split equally. There are 14 customers, 2 at each stall.

Where should a rational vendor locate?

Position 4 Beats Position 1

Compare market share when you play **1** vs. **4**, for each possible opponent location:

Opponent at	Your share at 1	Your share at 4
1	7	10
2	2	9
3	3	8
4	4	7
5	5	8
6	6	9
7	7	10

Position 4 strictly dominates position 1. By symmetry it also strictly dominates position 7.

No rational vendor will play 1 or 7. The effective game shrinks to positions $\{2, 3, 4, 5, 6\}$.

Now apply the same reasoning within the reduced game.

Position 4 Beats Position 2 (in the Reduced Game)

Opponent at	Your share at 2	Your share at 4
2	7	9
3	4	8
4	5	7
5	6	8
6	7	9

Position 4 strictly dominates position 2 in the reduced game, and by symmetry dominates position 6.

Round 2: Delete 2 and 6

The game shrinks further to $\{3, 4, 5\}$. Apply the reasoning once more.

Position 4 Beats Position 3 (in the Reduced Game)

Opponent at	Your share at 3	Your share at 4
3	7	8
4	6	7
5	7	8

Position 4 strictly dominates position 3 in this reduced game, and by symmetry dominates position 5.

Only position 4 survives. Both vendors locate at the centre.

This generalises: on any symmetric distribution of customers, rational vendors converge on the **median** location. The same logic applies to political candidates, petrol stations, fast food chains, and any competitor serving a distributed market.

Position 4 is also the unique Nash equilibrium of this game. Neither vendor can do better by moving, given the other is at the centre.

Traveler’s Dilemma

An airline loses two passengers’ identical antiques. It asks each to independently claim a value between \$200 and \$600. Both receive the *lower* claim, with adjustments:

- The lower claimer gets a \$200 bonus.
- The higher claimer gets a \$200 penalty.

Equal claims pay that amount to both.

If your opponent claims \$600 and you claim \$500, you receive $500 + 200 = \$700$, beating the \$600 you'd get by matching them.

So against an opponent who claims \$600, claiming \$500 is better.

But if the opponent claims \$400 or less, both \$500 and \$600 yield the same payoff (you both get the opponent's amount minus \$200). So \$500 only *weakly* dominates \$600, not strictly.

Suppose rational players won't claim \$600. The game reduces to $\{200, 300, 400, 500\}$. Now \$500 is weakly dominated by \$400 in the reduced game, by the same argument.

Each round knocks out the top claim. The only Nash equilibrium is for both to claim **\$200**.

The Problem with That Prediction

In experiments, most participants claim close to \$600. Average claims typically sit around \$500. Almost nobody claims \$200.

The step-by-step argument is valid at each stage, yet the conclusion is both worse for everyone and empirically rare.

This is one reason game theorists take experimental evidence seriously. Common knowledge of rationality is doing a lot of work in the iterated argument, and real players seem to stop the iteration early, or factor in that their opponents will too.

For Next Time

- We move from simultaneous games to **dynamic games**, where players move in sequence.
- This requires a new representation: the **game tree** (extensive form).
- And a new solution concept: **backward induction**.